

gstack Tutorial #5

From YouTube analytics to a re-engineered Short

CSR-RAP agent × Claude Opus 5.5 × HyperFrames — how the AGI Race V2 was made

Scope: V2 of the published AGI Race Short. Episode 2 (Mission: Impossible) is still in production and is not taught here.

September 2026

V2 raised structural quality by 27.9 %; whether viewers agree is still an open test

0.602 → 0.770

CSR-RAP B score

V1 proved the topic ($R \times A$). V2 adds the production system ($R \times P = 0.865$).

4.15 s

Hook complete

Punchline first, title second: the premise lands in 8.9 % of the Short.

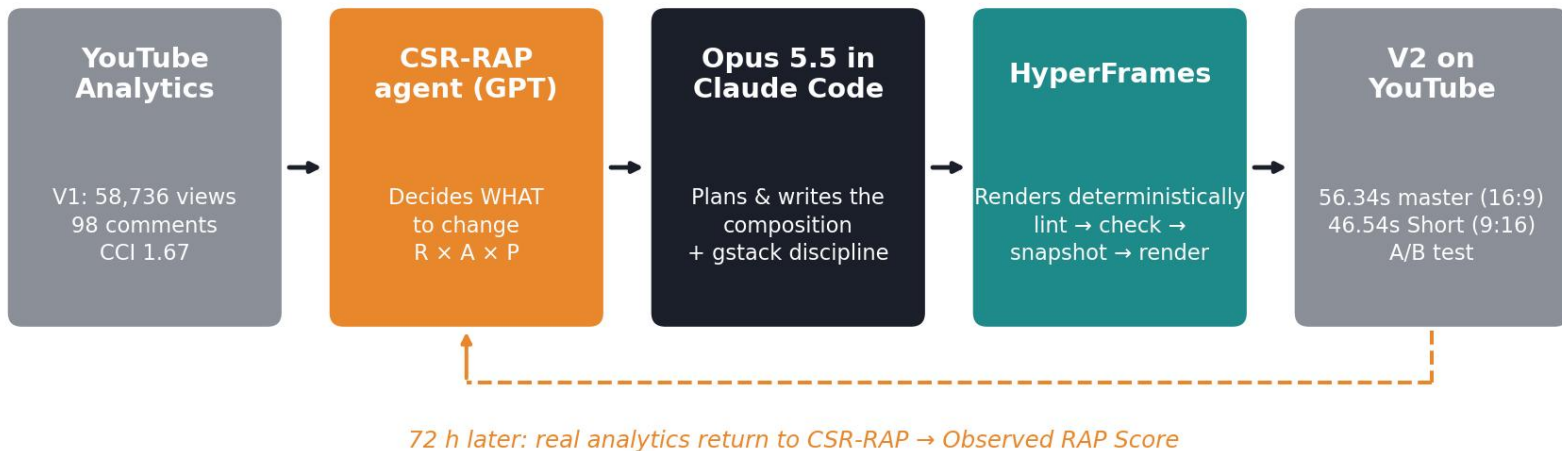
1.67 → ?

Comments per 1,000 views

Target > 2.5. Nothing is proven until 72 h of real V2 analytics return.

The leverage came from splitting the work across three systems with different jobs

Three layers, one loop: decide → build → render → measure



Decide (CSR-RAP) → build (Opus 5.5 + gstack) → render (HyperFrames) → measure, then loop.

Generative AI creates uncertain pixels; HyperFrames controls certain information

Must be exactly right → HyperFrames

- Timing of every beat, to the frame
- All on-screen text, captions, role tags
- 9:16 reframing per shot
- Freeze, black cut, end-card
- Audio cue placement

May vary → generative model / source footage

- Photoreal characters and environments
- Motion inside a shot
- Texture and lighting detail

Seek-safe test: if the renderer jumps straight to $t = 4.312$ s, is the frame still correct?

V2 needed almost nothing from the right column: its pixels already existed in V1.

The whole toolchain installs on Windows 11 in about 10 minutes; gstack takes one

1 Prerequisites

Git · Node.js ≥
22 · FFmpeg · Bun ≥
1.0 · Claude Code

```
winget install --id  
OpenJS.NodeJS -e
```

2 gstack (~1 min)

Paste the README one-liner
into Claude Code; test with
/office-hours

```
git clone ...  
garrytan/gstack.git  
~/.claude/skills/gstack  
&& ./setup
```

3 HyperFrames

Install the Claude Code
plugin; restart;
/hyperframes routes the job

```
claude plugin install  
hyperframes@hyperframes
```

4 Doctor first

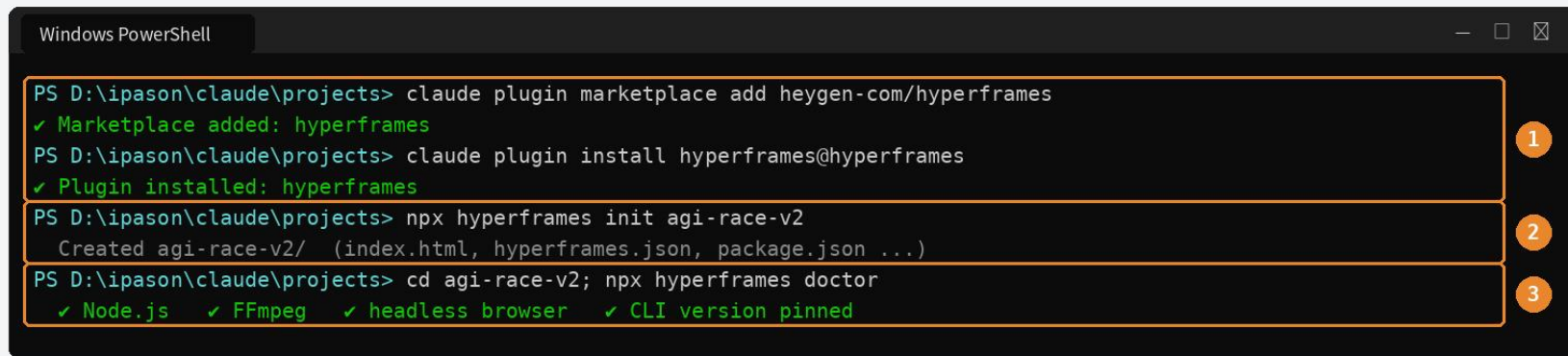
Check the machine before
designing anything; pin the
CLI version

```
npx hyperframes doctor
```

Pitfall we hit: 'claude' is not recognized → add %LOCALAPPDATA%\Microsoft\WinGet\Links to PATH and open a new PowerShell.

Two commands install HyperFrames; the doctor check runs before any design work

ILLUSTRATIVE MOCK-UP — replace with your own screenshot



```
Windows PowerShell
PS D:\ipason\claude\projects> claude plugin marketplace add heygen-com/hyperframes
✓ Marketplace added: hyperframes
PS D:\ipason\claude\projects> claude plugin install hyperframes@hyperframes
✓ Plugin installed: hyperframes
PS D:\ipason\claude\projects> npx hyperframes init agi-race-v2
Created agi-race-v2/ (index.html, hyperframes.json, package.json ...)
PS D:\ipason\claude\projects> cd agi-race-v2; npx hyperframes doctor
✓ Node.js ✓ FFmpeg ✓ headless browser ✓ CLI version pinned
```

- 1 Two commands install the HyperFrames plugin; restart Claude Code so /hyperframes appears.
- 2 init creates a composition project: plain HTML + data-start / data-duration timing.
- 3 Run doctor BEFORE designing — the Shot Playbook's rule: check the machine first.

Illustrative mock-up with annotations. Replace with your own screenshot.

V1 proved the topic works, but conversion did not follow the views

58,736

views (channel's first breakout)

98

comments

1.67

CCI = comments per 1,000 views

Layer	Score	Diagnosis
R Rules of Business	88	Cognitive compression: an AI race became a desert race
A Attributes of Industry	90	Hot topic, recognisable figures, strong Shorts fit
P Production capability	76	Weak causality; fails in 9:16
$B = R \times A \times P$	0.602	One weak layer caps the whole result

Measuring V1 first showed what to keep and exactly what broke on mobile

62.49 s

runtime

47

shots (46 hard cuts)

1.08 s

median shot; 20 of 46 under 1 s

0.95 s

mean shot in the fastest block, just before the climax

1.5 s + 0.84 s

near-freeze, then black

4

elements cut by a 9:16 crop: title, ensemble, final freeze, every subtitle

All values FFmpeg-measured by Opus 5.5 in Claude Code (Episode1_Visual_DNA.md).

An agent that only writes prompts for another generator is still gambling

“Entry gate: FAIL. Nothing has been generated yet... Numbers for frames nobody has generated would be invented.”

— Opus 5.5, text-to-video route, home PC

1

The gate worked

Opus refused to score footage that did not exist.

2

The dependency was wrong

The pipeline waited on the least controllable step: generated faces.

3

The pivot proved the control layer

C24 pilot: countdown correct at any seek time; black cut on frame 139; lint 0 errors; contrast 13/13.

4

...and showed its limit

C24 scored R 0.82, A 0.75: held back by placeholder characters, not structure.

Seven Shot Playbook rules turned one day's mistakes into permanent constraints

- 1 Route through /hyperframes; multi-scene → /general-video
- 2 Run doctor and read the timeline before designing
- 3 Search the catalog before building; reuse components
- 4 Seek-safe: no wall-clock logic, no Web Audio graphs, no conflicting fromTo
- 5 Holds over ≈ 4 s get one monotonic micro-motion, never a loop
- 6 Critical text in the centre third, so it survives 9:16
- 7 lint → check → snapshot → render → ffprobe, then human eyes

Renders are visually, not bit-for-bit, deterministic: 61 of 144 frames identical, the rest ≥ 50 dB PSNR.

The V2 brief pairs a change list from CSR-RAP with a keep list from the Visual DNA

Change (from CSR-RAP)

- Cold open on the strongest punchline
- Title slam; hook complete by ≈ 4 s
- Role tags replace name recognition
- Delete unintelligible lines
- 9:16 as its own composition
- End-card bridge to Episode 2

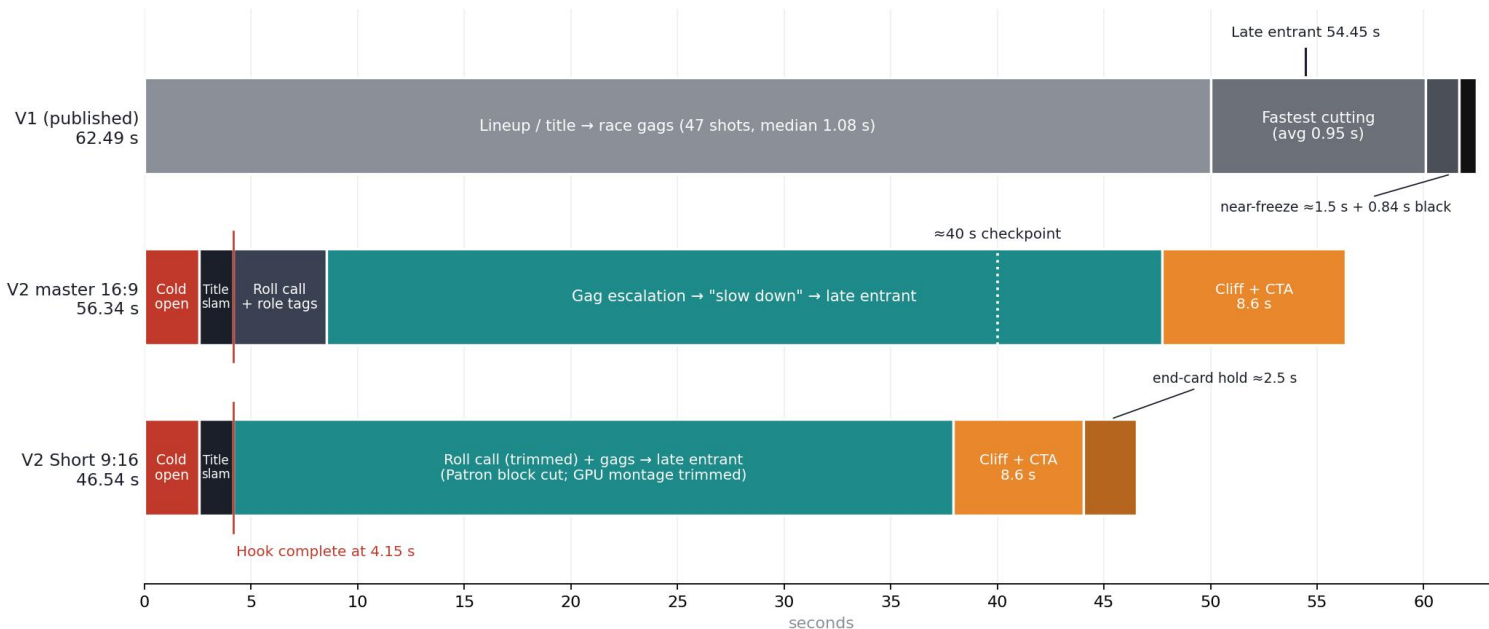
Keep (from Visual DNA)

- Fast grammar (median shot ≈ 1.08 s)
- Acceleration before the climax
- "Slow down" as the biggest close-up
- Unresolved jump
- Near-freeze $\rightarrow$ black ending
- Muted grade

Before the build: /office-hours $\rightarrow$ /plan-ceo-review (scope: HOLD, re-edit only) $\rightarrow$ /plan-eng-review. After it: /review.

V2 delivers spectacle, punchline and premise in the first 4.15 seconds

V1 vs V2: where the seconds went

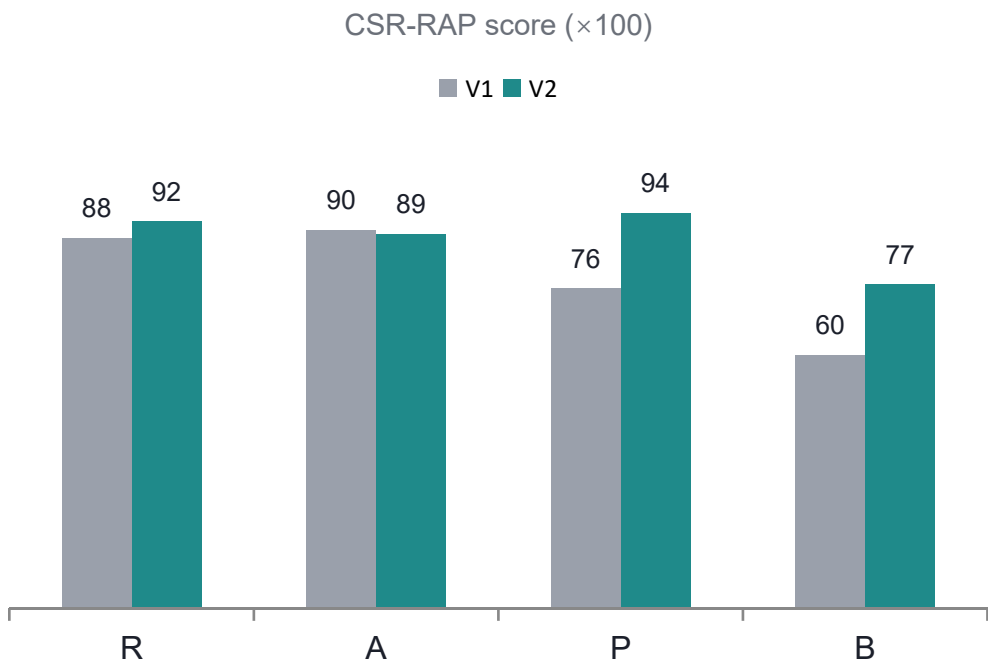


Anchors: V1 from Episode1_Visual_DNA (FFmpeg-measured); V2 from the V2 storyboard as quoted in the CSR-RAP V2 analysis. Inner beat boundaries not labelled are in the storyboard file.

Every V1 weakness the diagnosis named has a specific V2 answer

Metric	V1	V2	Assessment
Main runtime	62.49 s	56.34 s	−9.8 %
Short runtime	—	46.54 s	−25.5 % vs V1
Opening	Lineup / title	Strongest gag first	Clear upgrade
Hook complete	Slower	4.15 s	Stronger
Role comprehension	Relies on recognition	Explicit role tags	Major upgrade
9:16	Fails in several places	Shot-by-shot reframing	Major upgrade
Ending	Unresolved jump	Jump + Episode 2 bridge	Commercial upgrade
CTA system	Weak / generic	TO BE CONTINUED → EP 2 → SUBSCRIBE	Clear upgrade
Production	Video generation	Programmable HyperFrames re-edit	P upgrade

The strongest pair of layers moved from topic choice ($R \times A$) to production system ($R \times P$)



+27.9 %

structural quality, B 0.602 $\rightarrow$ 0.770

0.865

$R \times P$, now the strongest coupling, and it compounds across episodes

$\neq$ views

a design score, not a forecast; A stays provisional until real data arrives

Freeze both uploads for 72 hours and read three windows on the retention curve

The one question: does HyperFrames packaging turn the same content into better audience behaviour?

Window	Question	If it fails
0–4.15 s	Does the cold open stop the swipe?	Test another opening punchline
≈40 s	Does "slow down → late entrant" lift attention?	Tighten the escalation before it
Last 8.6 s	Does the end-card add subscribers or cause exits?	Cliff + no subscriber lift → hold 2.5 s → 1–1.5 s

Observed RAP Score (proposed): hook = stayed-to-watch % · retention = APV · engagement = CCI (target > 2.5) · subscribers per 1,000 views · vertical advantage = Short APV ÷ master APV

Four limits stand between this result and a proven one

1 No V2 analytics yet

Every V2 claim is structural until the 72-hour pull.

2 Scores are judgements

CSR-RAP rates design quality; it does not measure behaviour.

3 Generative layer untested

Episode 2's faces, hands and keys remain the open risk.

4 Real people, satirised

Role tags reduce reliance on likeness; review platform policy each release.

The same nine steps re-engineer any Short, not just this one

1

Export analytics; compute CCI

2

Diagnose with CSR-RAP

3

Measure: Visual DNA

4

Challenge: /office-hours

5

Brief Opus 5.5

6

Build + verify in HyperFrames

7

Re-score (design)

8

Upload, freeze, measure 72 h

9

Feed back into the next brief

Measure, don't re-edit

The next decision belongs to the viewers' data, not to another render.

Repo: [TUTORIAL.md](#) · [TUTORIAL.zh.md](#) · [starter/prompts](#) · [docs/SOURCES.md](#)

gstack: github.com/garrytan/gstack

HyperFrames: github.com/heygen-com/hyperframes